Sicilian, English and Italian Neural Machine Translation: an Automated Data Pipeline and Reproducible Baselines

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Abstract

Sicilian is a low-resource Romance language, with a literary tradition older than Italian, yet absent from every major machine-translation service. Building on Wdowiak's *Tradutturi Sicilianu* [Wdowiak, 2021], the first Sicilian neural machine translation (NMT) system, we rebuild the parallel-corpus pipeline with sentence-embedding extraction and alignment (PyMuPDF and LaBSE [Feng et al., 2022]), assemble a reproducible, deduplicated Sicilian–English dataset with a held-out *literary* test set, and port the paper's tokenization and desinence-biasing recipe to a Perl-free stack. We evaluate both from-scratch Sockeye baselines and the pretrained NLLB-200 model [NLLB Team, 2022] adapted with LoRA [Hu et al., 2021]. On our harder held-out test, Sicilian→English climbs from a 5.27 BLEU floor to 31.43 with a bidirectionally fine-tuned NLLB-1.3B—above Wdowiak's published baseline (29.1)—while the reverse direction nearly doubles (9.89→18.73). We further include **Italian** as a closely-related, higher-resource bridge language (`ita_Latn`): NLLB supports it↔scn natively, and we evaluate and fine-tune those directions on a frozen WikiMatrix it–scn split. The full pipeline, dataset assembly and evaluation are reproducible.

1 Introduction

Sicilian is a Romance language spoken daily by millions in Sicily, Calabria and Puglia, and in diaspora communities abroad. The Sicilian School of poets at the court of Frederick II created the first literary standard in Italy in the 13th century and inspired Dante; Sicilian thus emerged as a literary language *before* Italian. Despite this, it is absent from Google Translate, Bing, Yandex and DeepL—as are most regional languages.

The bottleneck was never the model but the *parallel text*: nobody had assembled it. The material exists, in the bilingual literary journal published for over forty years by Arba Sicula. Wdowiak's *Tradutturi Sicilianu* [Wdowiak, 2021] was the first Sicilian NMT system, trained on hand-aligned Arba Sicula text with a recipe tailored to the low-resource, morphologically rich setting. This work rebuilds that effort with maintained, mostly CPU-only tooling and pushes quality with modern pretrained models.

Contributions.

- An automated extraction and alignment pipeline for the *Arba Sicula* PDFs (PyMuPDF and LaBSE [Feng et al., 2022]), replacing the legacy pdflatex+hunalign flow, with thresholds tuned against Wdowiak's hand-aligned gold.
- A reproducible, deduplicated, provenance-tracked dataset with a held-out literary test set (not FLORES), and a BLEU/chrF evaluation harness.
- The paper's "lever B" recipe (Napizia tokenizer + desinence-biased subwords) ported to a Perl-free pipeline, and a linguistic-feature lever (lemma source factors).
- NLLB-200 zero-shot and LoRA fine-tuning that beats the published baseline on our harder test set, with a bidirectional adapter that also rescues the reverse direction.

2 Background

NMT models translation as a conditional language model: an encoder reads the source $x = (x_1, \ldots, x_n)$ and a decoder factorises the target left to right, $p_\theta(y \mid x) = \prod_t p_\theta(y_t \mid y_{<t}, x)$, trained by maximising the log-likelihood of a parallel corpus and decoded with beam search. The dominant parameterisation is the **Transformer** [Vaswani et al., 2017], which replaces recurrence with scaled dot-product attention, $\mathrm{Attention}(Q, K, V) = \mathrm{softmax}(QK^\top/\sqrt{d_k})V$, run in parallel heads.

With little data the default Transformer overfits; the low-resource recipe of Sennrich and Zhang [2019] prescribes a smaller, heavily regularised model with high dropout [Srivastava et al., 2014] and small subword vocabularies. **Subword segmentation** via byte-pair encoding [Sennrich et al., 2016] gives an open vocabulary and shares statistics across word forms—essential for a morphologically rich language. Further gains come from **linguistic input features** [Sennrich and Haddow, 2016], **back-translation** of monolingual data [Sennrich et al., 2015], and **multilingual** training with a target-language token [Johnson et al., 2016] that enables bridging through related high-resource languages [Fan et al., 2020, Arivazhagan et al., 2019]. Low-resource MT remains hard for exactly these reasons [Koehn and Knowles, 2017].

3 Data Pipeline

3.1 Extraction from PDFs

We extract clean text from the Arba Sicula issue PDFs with PyMuPDF (no pdflatex round-trip). Each page is assigned a language from its share of Sicilian stop-words and diacritics; the journal sets Sicilian and English on facing pages, giving a natural even/odd prior, so we form facing candidate pairs. Pages are segmented into sentences with `pysbd` (per language), filtering "furniture" (all-caps headers, page numbers, captions). The procedure recovers parallel pages the manual selection had skipped.

3.2 Sentence alignment

We embed every sentence with LaBSE [Feng et al., 2022], a language-agnostic dual encoder, and build the cosine-similarity matrix $S_{ij} = \langle u_i, v_j \rangle / (\|u_i\| \|v_j\|)$. A facing pair is confirmed only if its mean best-match similarity exceeds τ_{page}; sentences are then aligned by a monotonic dynamic program over S allowing 1–1, 1–2 and 2–1 moves with a null penalty, in the spirit of Vecalign [Thompson and Koehn, 2019] but without a bilingual dictionary or hunalign. The thresholds $(\tau_{\text{page}}, \tau_{\text{sent}}) = (0.50, 0.40)$ were tuned against Wdowiak's hand-aligned gold (issues AS41–42); the pipeline recovers 14,245 sentence pairs from 44 issues, comparable to his manual alignment. Crucially, the *unaligned* English is retained as an in-domain monolingual pool for back-translation.

3.3 Additional sources

We add the author's curated "Good-Sicilian-in-NLLB" set—LASER3-mined sentences [Heffernan et al., 2022, NLLB Team, 2022] rescored and filtered for quality and Corsican contamination—a small curated "Good-Sicilian-from-WikiMatrix" set, and the WikiMatrix it–scn subset [Schwenk et al., 2021] as an auxiliary Italian bridge.

3.4 Unified dataset

Sources are cleaned, deduplicated across one another, and split deterministically. A 1,000/1,000 valid/test split is *frozen* once, held out from Arba Sicula (a literary standard, not FLORES), so every later result stays comparable (Table 1).

Table 1: Unified dataset.

split	pairs	source
train	27,386	Arba Sicula + NLLB
valid	1,000	Arba Sicula
test	1,000	Arba Sicula (literary)
it–scn (aux)	11,389	WikiMatrix

4 Tokenization and Subwords

We re-implement the Napizia tokenizer in pure Python (verified identical to the original Perl): it ASCII-folds accents and *uncontracts* Sicilian (e.g. ô→*a lu*, dû→*di lu*), cutting sparsity. "Lever B" then biases the joint scn–en BPE [Sennrich et al., 2016] toward the textbook *desinences* catalogued in the resources Wdowiak assembled to standardise Sicilian—Dieli's *Sicilian Vocabulary*, Cipolla's *Mparamu lu sicilianu* [Cipolla, 2013], Bonner's grammar [Bonner, 2001], and the *Chiù dâ Palora* dictionary—so morphological endings survive as units (Wdowiak's example: *Carinisi are dogs!* segments to `car@@ in@@ isi are dogs !`). This cut the vocabulary from 14,000 to 2,000 units and produced his largest BLEU gains. BPE-dropout [Provilkov et al., 2020] adds subword regularisation by sampling segmentations during training.

5 Models

The from-scratch baseline is a Sockeye-3 [Hieber et al., 2017] small Transformer (3 layers, model size 256, 4 heads, feed-forward 1024, weight tying, dropout 0.4, joint BPE ≈4k; ∼6.6M parameters), trained on CPU. On top of it we stack lever B (tokenization + desinences), more parallel text, and "lever D" linguistic input features [Sennrich and Haddow, 2016]: a lemma source factor summed into the source embedding, $e_j = E_{\text{word}}(w_j) + E_{\text{lemma}}(\ell_j)$, whose small (∼900) vocabulary generalises across inflected forms.

The pretrained model is NLLB-200 [NLLB Team, 2022], a massively multilingual encoder–decoder that supports Sicilian (`scn_Latn`) and is steered by forcing the first decoder token. We adapt the 1.3B variant with LoRA [Hu et al., 2021]: freezing W_0 and learning a low-rank update $W = W_0 + \frac{\alpha}{r}BA$ (r=32, α=64) on the attention projections, with the base in fp16 and the adapters in fp32. Training *both* directions at once (scn→en and en→scn) yields a single adapter that serves both and rescues the weak reverse direction.

Table 2: Our models, our test set (scn→en).

model	BLEU	chrF
floor (copy source)	5.27	25.40
Sockeye baseline	5.54	28.28
+ lever B (tok. + desinences)	7.24	29.52
+ more data	9.79	33.82
+ lever D (lemma source factors)	10.85	35.22
NLLB-200 1.3B, zero-shot (raw)	29.02	55.23
NLLB-200 1.3B, LoRA fine-tuned, scn→en only	31.16	56.79
NLLB-200 1.3B, LoRA *bidirectional*	**31.43**	**56.94**

6 Evaluation

We score with sacreBLEU [Post, 2018]—BLEU [Papineni et al., 2002] (`tok:13a`) and chrF [Popović, 2015]—on the frozen literary test set rather than the NLLB/FLORES set: the FLORES Sicilian uses the post-2017 orthography and a different (non-literary) domain, so it is not representative of the Arba Sicula material. The Sockeye rows are scored in tokenized space (the same space as the upstream system; the raw-text floor is 5.27 BLEU), while the NLLB rows are scored on raw text.

7 Results

On the frozen held-out test set (1,000 literary pairs), Sicilian→English results are reported in Table 2. The Sockeye levers stack from 5.54 to 10.85 BLEU on CPU alone; the pretrained NLLB-1.3B wins decisively even zero-shot (29.02), and LoRA fine-tuning lifts it to 31.43—above Wdowiak's published baseline (29.1) on our harder test.

In the reverse direction (en→scn), the same bidirectional adapter lifts NLLB-1.3B from 9.89 (zero-shot) to 18.73 BLEU / 49.96 chrF, nearly doubling the weak direction at no cost to scn→en and giving a usable two-way model. This is still below the published en→scn baseline (25.1); back-translation is the next lever.

Italian directions (it↔scn). Italian is a closely related, much higher-resource Romance language, and NLLB supports it↔scn natively. We hold out a frozen 1,000-pair it–scn test split from the WikiMatrix data, evaluate NLLB-1.3B zero-shot, then fine-tune *all four* directions jointly (scn↔en and it↔scn) into a single multilingual adapter, a bridge in the sense of Johnson et al. [2016] and Fan et al. [2020] (Table 3). Italian is by far the easiest pair—scn→it reaches **43.5** BLEU and it→scn climbs 17.6 → **27.0**—as expected

Table 3: Trilingual model: all directions, zero-shot vs. the multilingual fine-tune (BLEU / chrF, frozen test sets).

direction	zero-shot	multilingual FT
scn→en	29.02 / 55.23	30.75 / 56.57
en→scn	9.89 / 41.12	17.49 / 48.78
it→scn	17.61 / 44.77	**26.97** / **51.69**
scn→it	42.20 / 59.04	**43.47** / **59.79**

for a close, high-resource neighbour on in-domain WikiMatrix text. The multilingual adapter costs a little on scn↔en (30.8/17.5 vs the dedicated 31.4/18.7), so one can ship either the trilingual all-rounder or the specialised pair.

Relation to the state of the art. We do *not* beat the state of the art. Wdowiak's **reverse-training** system reports, on his own in-domain hand-selected test set, En→Sc/Sc→En BLEU of 45.1/48.6 and It→Sc/Sc→It of 61.4/62.9—well ahead of our 18.7/31.4 and 27.0/43.5 on every direction (his earlier marks were 25.1/29.1 baseline, 35.0/36.8 augmented). The numbers are *not* a head-to-head (different test sets), but the gap is too large to be explained by that alone: on Sc→en we only clear his *baseline* (31.4 vs 29.1), on a harder test, and trail elsewhere. Crucially, his reverse-training builds a *custom pretrained model* from tens of millions of pairs (forward-translation, then back-translation, in three stages) before fine-tuning on *Arba Sicula*—a pipeline infeasible to reproduce on a single GPU. NLLB-200, however, already provides that massive multilingual pretraining: our fine-tune is the analogue of his stage-3. Adding back-translation and multilingual training (in progress) closes the remaining *method* gap; the likely *residual* is then **data**—his hand-curated, standardised corpus versus our automatic extraction—which argues for combining the two rather than competing.

8 Conclusion and Future Work

We presented a reproducible, Perl-free path from Arba Sicula PDFs to a Sicilian–English NMT system: an automated extraction-and-alignment pipeline, the paper's tokenization and desinence recipe with an added linguistic-feature lever, and NLLB-200 adapted with LoRA. On our harder literary test set the best model reaches 31.43 BLEU (scn→en), above the published baseline but well below Wdowiak's reverse-training state of the art. Ongoing work applies the feasible parts of his recipe on top of NLLB—**back-translation** from the in-domain monolingual English harvested from unaligned text, and **multilingual** Italian bridging through the it–scn data [Fan et al., 2020]—to isolate how much of the gap is *method* versus *data*. If, with the method matched, our automatically-extracted corpus still trails his hand-curated one, the difference is the data: a concrete case for combining his corpus with our pipeline. We also release **serving** (a local web app and a Telegram bot, a FastAPI service over the NLLB adapter). All code, the data pipeline and the evaluation harness are released for reproducibility.

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